3.31 Dedekind Domains

We develop the theory of Dedekind Domains. These occur both as number fields and regular affine curves (over any field). The important results are the finiteness of integral closure in a finite extension, and the ramification formula of a prime ideal. In the case of regular curves over a finite field further results are needed, which we prove later in the context of affine algebras.

Proposition 3.31.1
Let A be a Noetherian integral domain of dimension 1. Then the following are equivalent

a) A is integrally closed
b) $A_\mathfrak{p}$ is integrally closed for every prime ideal $\mathfrak{p}$
c) $A_\mathfrak{p}$ is a discrete valuation ring for every non-zero prime ideal $\mathfrak{p}$

*In this case we say that A is a **Dedekind Domain**.*

Proof. Let $\mathfrak{p}$ be a non-zero prime ideal of A. Then $A_\mathfrak{p}$ is Noetherian (3.14.7), local, and an integral domain as a subring of $\mathrm{Frac}(A)$. Moreover $\dim A_\mathfrak{p} = \mathrm{ht}(\mathfrak{p}) \leq \dim A = 1$, and $(0) \subsetneq \mathfrak{p}$ so $\dim A_\mathfrak{p} = \mathrm{ht}(\mathfrak{p}) = 1$.

a) $\iff$ b) This is (3.23.10).

b) $\implies$ c) By the above $A_\mathfrak{p}$ is a Noetherian local domain of dimension 1, and it is integrally closed by b). Hence it is a discrete valuation ring by (3.30.2) g).

c) $\implies$ b) For $\mathfrak{p}$ non-zero, $A_\mathfrak{p}$ is a discrete valuation ring, hence integrally closed by (3.30.2) g). For $\mathfrak{p} = (0)$ we have $A_{(0)} = \mathrm{Frac}(A)$, a field, which is automatically integrally closed. ∎

Corollary 3.31.2
Every principal ideal domain which is not a field is a Dedekind Domain. For example $k[X]$ and $\mathbb{Z}$.

Proof. We check the hypotheses of (3.31.1), taking condition a). Let A be a PID which is not a field.

- A is a Noetherian integral domain (3.15.2).
- A has dimension 1: every non-zero prime ideal of A is maximal (3.16.23), so $\dim A \leq 1$; and as A is not a field it has a non-zero maximal ideal $\mathfrak{m} \supsetneq (0)$, so $\dim A \geq 1$.
- A is integrally closed, as A is a UFD (3.16.23) and a UFD is integrally closed (3.23.12).

∎

Corollary 3.31.3
Every discrete valuation ring is a Dedekind Domain.

Proof. A discrete valuation ring A is a Noetherian integrally closed domain of dimension 1 (and local) (3.30.2) g), which is condition a) of (3.31.1). ∎

The following is useful in the theory of algebraic curves and function fields.

Proposition 3.31.4 (Correspondence between Prime Ideals and Valuation Rings)
Let A be a Dedekind domain with quotient field K. Then there is a bijection

$$\{\mathfrak{p} \lhd A \mid \mathfrak{p} \text{ non-zero prime}\} \longleftrightarrow \{R \text{ local ring} \mid A \subseteq R \subsetneq K\}$$
$$\mathfrak{p} \longrightarrow A_\mathfrak{p}$$
$$\mathfrak{m}_R \cap A \longleftarrow R$$

and each such local ring R is a discrete valuation ring.

Proof. Let $\mathfrak{p}$ be a non-zero prime ideal. Then $A_\mathfrak{p}$ is a discrete valuation ring by (3.31.1), in particular local with maximal ideal $\mathfrak{p}A_\mathfrak{p}$, and $A \subseteq A_\mathfrak{p} \subseteq K$. Also $A_\mathfrak{p} \neq K$: if $0 \neq a \in \mathfrak{p}$ and $a^{-1} = x/s$ with $x \in A$, $s \in A \setminus \mathfrak{p}$, then $s = ax \in \mathfrak{p}$, a contradiction. So $\mathfrak{p} \mapsto A_\mathfrak{p}$ is well-defined.

Next, $\mathfrak{m}_{A_\mathfrak{p}} \cap A = \mathfrak{p}$. Indeed if $a = x/s$ with $x \in \mathfrak{p}$ and $s \in A \setminus \mathfrak{p}$ then $as = x \in \mathfrak{p}$, so $a \in \mathfrak{p}$ as $\mathfrak{p}$ is prime, and the other inclusion is clear. In particular $\mathfrak{p} \mapsto A_\mathfrak{p}$ is injective.

Conversely let R be a local ring with $A \subseteq R \subsetneq K$, and put $\mathfrak{p} := \mathfrak{m}_R \cap A$, a prime ideal of A. As R is not a field (a field $R \supseteq A$ contains $\mathrm{Frac}(A) = K$) there is $0 \neq x/y \in \mathfrak{m}_R$ with $x, y \in A$, whence $0 \neq x = y \cdot (x/y) \in \mathfrak{m}_R \cap A = \mathfrak{p}$. So $\mathfrak{p}$ is non-zero. If $x \in A \setminus \mathfrak{p}$ then $x \in R \setminus \mathfrak{m}_R$, so $x \in R^\star$; hence $A_\mathfrak{p} \subseteq R$ and $\mathfrak{p}A_\mathfrak{p} \subseteq \mathfrak{m}_R$, that is R dominates $A_\mathfrak{p}$. As $A_\mathfrak{p}$ is a discrete valuation ring it is a valuation ring (3.30.2), so it is maximal under domination (3.24.13), and $R = A_\mathfrak{p}$.

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This shows $\mathfrak{p} \mapsto A_{\mathfrak{p}}$ is surjective with inverse $R \mapsto \mathfrak{m}_R \cap A$, and every R is of the form $A_{\mathfrak{p}}$, hence a discrete valuation ring. ∎

Remark 3.31.5
The proof of injectivity works for any integral domain A and any prime ideal $\mathfrak{p}$: $\mathfrak{p} \mapsto A_{\mathfrak{p}}$ has left inverse $R \mapsto \mathfrak{m}_R \cap A$. It is surjectivity onto the local rings between A and K that needs A to be Dedekind (to make each $A_{\mathfrak{p}}$ a valuation ring). It fails for $A = k[t^2, t^3]$, where $k[t]_{(t)}$ is a local ring between A and $K = k(t)$ with $\mathfrak{m}_R \cap A = (t^2, t^3)$, the same centre as $A_{(t^2,t^3)}$ but a different ring.

Proposition 3.31.6
Let A be a Dedekind domain. Then every fractional ideal is invertible.

Proof. Let I be a fractional ideal. As A is Noetherian ((3.31.1)), I is finitely generated (3.29.2). So by (3.29.11) it is sufficient to show that $I_{\mathfrak{m}}$ is invertible for every maximal ideal $\mathfrak{m}$. As $\dim A = 1$ the ring A is not a field, so $\mathfrak{m}$ is non-zero and $A_{\mathfrak{m}}$ is a discrete valuation ring by (3.31.1). Now $I_{\mathfrak{m}}$ is a fractional ideal of $A_{\mathfrak{m}}$ (3.29.9), so it is invertible by (3.30.7). ∎

Lemma 3.31.7
Let A be a Dedekind domain and $\mathfrak{a}$ a non-zero ideal. Then there is a well-defined group homomorphism

$$(\mathfrak{a})^- : \mathbb{Z} \longrightarrow \mathcal{I}_A$$

mapping 1 to $\mathfrak{a}$.

Proof. By (3.31.6) the monoid $\mathcal{I}_A$ is a group, and therefore the first homomorphism exists by (3.2.6). ∎

Lemma 3.31.8
Let A be a Dedekind domain, $\mathfrak{p}$, $\mathfrak{q}$ non-zero prime ideals and $k \in \mathbb{Z}$. Then

$$\left(\mathfrak{p}^k\right)_{\mathfrak{q}} = \begin{cases} A_{\mathfrak{q}} & \mathfrak{p} \neq \mathfrak{q} \\ \left(\mathfrak{q}A_{\mathfrak{q}}\right)^k & \mathfrak{p} = \mathfrak{q} \end{cases}$$

Proof. When $k = 0$ then by definition $\mathfrak{p}^k = A$ and $\left(\mathfrak{q}A_{\mathfrak{q}}\right)^k = A_{\mathfrak{q}}$ and so we are done.

We suppose that $k = 1$. If $\mathfrak{p} \neq \mathfrak{q}$ then as $\dim A = 1$ we have $\mathfrak{p} \not\subseteq \mathfrak{q}$. Therefore by definition $\mathfrak{p}_{\mathfrak{q}}$ contains 1 and equals $A_{\mathfrak{q}}$ as required. The case $\mathfrak{p} = \mathfrak{q}$ follows from (3.29.8).

By (3.29.9) and (3.31.7) both left and right hand sides define group homomorphisms $\mathbb{Z} \to \mathcal{I}_{A_{\mathfrak{q}}}$ which agree at 1 and are therefore equal. ∎

Proposition 3.31.9 (Unique Factorisation)
Let A be a Dedekind domain and $\mathcal{P}$ the set of non-zero prime ideals of A. There is a well-defined abelian group isomorphism

$$\bigoplus_{\mathfrak{p} \in \mathcal{P}} \mathbb{Z} \xrightarrow{\sim} \mathcal{I}_A$$
$$(e_{\mathfrak{p}}) \longrightarrow \prod_{\mathfrak{p}: e_{\mathfrak{p}} \neq 0} \mathfrak{p}^{e_{\mathfrak{p}}}$$

Write $\left(v_{\mathfrak{p}}\right)_{\mathfrak{p} \in \mathcal{P}}$ for the inverse, then it is given explicitly by

$$v_{\mathfrak{p}}(I) = v_{\mathfrak{p}}\left(I_{\mathfrak{p}}\right)$$

where on the right $v_{\mathfrak{p}}$ is the valuation of the discrete valuation ring $A_{\mathfrak{p}}$ (3.30.7). Further a fractional ideal I is contained in A if and only if $v_{\mathfrak{p}}(I) \geq 0$ for all $\mathfrak{p} \in \mathcal{P}$.

Proof. The homomorphisms exists by (3.31.7) and the universal property of the sum of abelian groups.

Let I be a fractional ideal and define $v_{\mathfrak{p}}(I) := v_{\mathfrak{p}}\left(I_{\mathfrak{p}}\right)$ using the notation of (3.30.7). This determines a group homomorphism $v_{\mathfrak{p}} : \mathcal{I}_A \to \mathbb{Z}$ by (3.29.9) b). We claim for every I that $v_{\mathfrak{p}}(I)$ is zero for all but finitely many $\mathfrak{p} \in \mathcal{P}$. As $I = (a)^{-1}\mathfrak{a}$ for some $0 \neq a \in A$ and non-zero ideal $\mathfrak{a}$ of A it is sufficient by multiplicativity to demonstrate this for non-zero ideals $\mathfrak{a}$. If $\mathfrak{a} \not\subset \mathfrak{p}$ then $\mathfrak{a}_{\mathfrak{p}} = A_{\mathfrak{p}}$ (as $\mathfrak{a}$ contains an element of $A \setminus \mathfrak{p}$) which has valuation 0. Therefore we may assume that $\mathfrak{a} \subset \mathfrak{p}$. However by (3.21.6) there are only finitely many such primes, since every prime containing $\mathfrak{a}$ must be minimal over $\mathfrak{a}$ (as $\mathfrak{a} \neq 0$ and $\dim A = 1$).

Observe by (3.31.8) that for distinct $\mathfrak{p}_1, ..., \mathfrak{p}_n \in \mathcal{P}$ we have the identity

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$$\left(\mathfrak{p}_1^{r_1}\ldots\mathfrak{p}_n^{r_n}\right)_{\mathfrak{q}} = \begin{cases} \left(\mathfrak{p}_k A_{\mathfrak{q}}\right)^{r_k} & \mathfrak{q} = \mathfrak{p}_k \\ A_{\mathfrak{q}} & \mathfrak{q} \neq \mathfrak{p}_k \text{ for all } k = 1\ldots n \end{cases}$$

By the first claim there is a well-defined homomorphism $\mathcal{I}_A \to \bigoplus_{\mathfrak{p}\in\mathcal{P}} \mathbb{Z}$. We claim that this is a right inverse for the given map, that is to say

$$\prod_{\mathfrak{p}:v_{\mathfrak{p}}(I)\neq 0} \mathfrak{p}^{v_{\mathfrak{p}}(I)} = I$$

Denote the left hand side by I'. When $v_{\mathfrak{p}}(I) \neq 0$ then by the identity above we have, using (3.30.7) for the final equality,

$$I'_{\mathfrak{p}} = \left(\mathfrak{p}A_{\mathfrak{p}}\right)^{v_{\mathfrak{p}}(I)} = \left(\mathfrak{p}A_{\mathfrak{p}}\right)^{v_{\mathfrak{p}}\left(I_{\mathfrak{p}}\right)} = I_{\mathfrak{p}}$$

Similarly if $v_{\mathfrak{p}}(I) = 0$ by (3.30.7)

$$I'_{\mathfrak{p}} = A_{\mathfrak{p}} = I_{\mathfrak{p}}$$

We conclude by (3.29.10) (every maximal ideal of A lies in $\mathcal{P}$, as A is not a field) that $I = I'$ as required. Finally the relation above also shows that this is a left inverse.

For the final statement by (3.29.10) we have $I \subset A \iff I_{\mathfrak{p}} \subset A_{\mathfrak{p}}$ for all $\mathfrak{p} \in \mathcal{P}$, then we may use the equivalent statement over DVRs (3.30.7). ∎

Proposition 3.31.10
Let A be a Dedekind domain with I, J fractional ideals. Then the following are equivalent

a) $I \subset J$
b) J divides I (that is there exists a non-zero ideal $\mathfrak{a}$ such that $\mathfrak{a}J = I$)
c) $v_{\mathfrak{p}}(I) \geq v_{\mathfrak{p}}(J)$ for all $\mathfrak{p} \in \mathcal{P}$

Proof. a) $\iff$ c) By (3.29.10) we have $I \subset J \iff I_{\mathfrak{p}} \subset J_{\mathfrak{p}}$ for all $\mathfrak{p} \in \mathcal{P}$ (if $I_{\mathfrak{m}} \subset J_{\mathfrak{m}}$ for every maximal ideal $\mathfrak{m}$ then $I = \bigcap_{\mathfrak{m}} I_{\mathfrak{m}} \subset \bigcap_{\mathfrak{m}} J_{\mathfrak{m}} = J$, and the maximal ideals are exactly the elements of $\mathcal{P}$). Then we may use the corresponding statement for DVRs (3.30.8).

b) $\implies$ c) Follows because $v_{\mathfrak{p}}(I) = v_{\mathfrak{p}}(\mathfrak{a}) + v_{\mathfrak{p}}(J)$ and $v_{\mathfrak{p}}(\mathfrak{a}) \geq 0$ (3.31.9).

c) $\implies$ b) We may define $\mathfrak{a} = \prod \mathfrak{p}^{v_{\mathfrak{p}}(I)-v_{\mathfrak{p}}(J)}$ (a finite product, as $v_{\mathfrak{p}}(I) = v_{\mathfrak{p}}(J) = 0$ for all but finitely many $\mathfrak{p}$ (3.31.9)) which is contained in A and by multiplicativity $v_{\mathfrak{p}}(\mathfrak{a}J) = v_{\mathfrak{p}}(I)$ for all $\mathfrak{p} \in \mathcal{P}$ whence $\mathfrak{a}J = I$ by the correspondence in (3.31.9). ∎

Proposition 3.31.11 (Integral Closure is a Dedekind Domain)
Let L/K be a finite separable extension and $A \subset K$ a subring which is a Dedekind domain with quotient field K. Let B be the integral closure of A in L, then B is a finite A-module and a Dedekind domain.

If in addition A is a principal ideal domain (e.g. a discrete valuation ring) then B is a free A-module of rank $[L : K]$.

Moreover $B \otimes_A K \cong L$, via $b \otimes x \mapsto bx$ (regarding $B, K \subseteq L$).

Proof. From (3.23.24) we know B is integrally closed and $\dim B = \dim A = 1$. As A is Dedekind it is Noetherian and integrally closed, so (3.23.26) shows that B is a finite A-module (hence Noetherian), and free of rank $[L : K]$ if A is a principal ideal domain. So B is a Noetherian integrally closed domain of dimension 1, that is a Dedekind domain (3.31.1).

Base change to K. Writing $S := A \setminus \{0\}$, (3.5.31) gives an isomorphism $B \otimes_A K \cong S^{-1}B$, $b \otimes x \mapsto bx$, where $S^{-1}B$ is naturally a subring of L (as $S \subset B \setminus \{0\}$ too). By (3.23.15) (applied to the integral map $A \hookrightarrow B$, and S), $K = S^{-1}A \to S^{-1}B$ is integral; as $S^{-1}B$ is an integral domain (a subring of L) and K is a field, (3.23.14) gives that $S^{-1}B$ is itself a field. Being a field containing B, $S^{-1}B$ contains $\mathrm{Frac}(B) = L$; but $S^{-1}B \subseteq L$ already; so $S^{-1}B = L$. ∎

Definition 3.31.12 (AKLB Situation)
Let L/K be a finite field extension, $A \subset K$, $B \subset L$ Dedekind domains for which $A \subset B$ is integral (equivalently B is the integral closure of A in L), $\mathrm{Frac}(A) = K$ and $\mathrm{Frac}(B) = L$. We refer to this situation as "AKLB".

Lemma 3.31.13
Using the notation in (3.31.12). Let $\mathfrak{p}$ be a non-zero prime ideal of A, and $\mathfrak{b}$ a proper ideal of B. Then the following are equivalent

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a) $\mathfrak{p} \subset \mathfrak{b}$
b) $\mathfrak{b}$ *divides* $\mathfrak{p}B$
c) $\mathfrak{b} \cap A = \mathfrak{p}$

and in this case we say that $\mathfrak{b}$ *divides* $\mathfrak{p}$ *and write* $\mathfrak{b} \mid \mathfrak{p}$.

In this case $B/\mathfrak{b}$ *is naturally an* $A/\mathfrak{p}$*-module via the commutative diagram*

$$\begin{array}{ccc} A & \hookrightarrow & B \\ \downarrow & & \downarrow \\ A/\mathfrak{p} & \hookrightarrow & B/\mathfrak{b} \end{array}$$

Proof. a) $\iff$ b) follows from (3.31.10)

a) $\implies$ c) Trivially $\mathfrak{p} \subset \mathfrak{b} \cap A$. As $1 \notin \mathfrak{b}$ then $\mathfrak{b} \cap A$ is a proper ideal, and therefore equal to $\mathfrak{p}$ by maximality (a non-zero prime of the Dedekind domain A is maximal, as $\dim A = 1$).

c) $\implies$ a) Immediate. ∎

Proposition 3.31.14
Using the notation in (3.31.12). Let $\mathfrak{p}$ *be a non-zero prime ideal of* A*. There exists at least one prime ideal* $\mathfrak{q} \mid \mathfrak{p}$*, and only finitely many. For every such prime ideal define the integers* $e_{\mathfrak{q}/\mathfrak{p}} := v_{\mathfrak{q}}(\mathfrak{p}B)$*, then*

$$\mathfrak{p}B = \prod_{\mathfrak{q}|\mathfrak{p}} \mathfrak{q}^{e_{\mathfrak{q}/\mathfrak{p}}} = \bigcap_{\mathfrak{q}|\mathfrak{p}} \mathfrak{q}^{e_{\mathfrak{q}/\mathfrak{p}}}$$

Proof. The existence of a prime $\mathfrak{q}$ follows from the Lying Over property of integral extensions (3.23.17). The factorisation exists by (3.31.9) which also shows that there are only finitely many divisors of $\mathfrak{p}$, and the second equality holds by the Chinese Remainder Theorem (3.3.63). ∎

Definition 3.31.15 (Ramification Index / Inertial Degree)
Using the notation in (3.31.12), suppose that $\mathfrak{q} \mid \mathfrak{p}$*. Then the* ***ramification index*** *is the integer*

$$e_{\mathfrak{q}/\mathfrak{p}} = v_{\mathfrak{q}}(\mathfrak{p}B)$$

which is positive: $\mathfrak{q}$ *divides* $\mathfrak{p}B$ *(3.31.13), so* $\mathfrak{p}B = \mathfrak{a}\mathfrak{q}$ *for an ideal* $\mathfrak{a}$ *of* B *and* $v_{\mathfrak{q}}(\mathfrak{p}B) = v_{\mathfrak{q}}(\mathfrak{a}) + 1 \geq 1$ *(3.31.9). Similarly define the* ***inertial degree***

$$f_{\mathfrak{q}/\mathfrak{p}} = [B/\mathfrak{q} : A/\mathfrak{p}]$$

which we presently show to be finite.

Lemma 3.31.16 (Removing a Common Factor in a PID)
Let A *be a PID and* $\lambda_1, ..., \lambda_n \in A$ *not all zero. Then there are* $0 \neq d \in A$ *and* $\mu_1, ..., \mu_n \in A$ *with* $\lambda_i = d\mu_i$ *for every* i *and* $(\mu_1, ..., \mu_n) = A$*. If* A *is moreover a local ring with maximal ideal* $\mathfrak{m}$ *then* $\mu_i \notin \mathfrak{m}$ *for some* i*.*

Proof. As A is a PID, $(\lambda_1, ..., \lambda_n) = (d)$ for some d, and $d \neq 0$ as the λ_i are not all zero. Each $\lambda_i \in (d)$, so $\lambda_i = d\mu_i$ for some $\mu_i \in A$. Also $d = \sum_i a_i\lambda_i = d\sum_i a_i\mu_i$ for some $a_i \in A$, so $\sum_i a_i\mu_i = 1$ as A is a domain and $d \neq 0$, that is $(\mu_1, ..., \mu_n) = A$. If A is local then a proper ideal lies in $\mathfrak{m}$, so the μ_i cannot all lie in $\mathfrak{m}$. ∎

Remark 3.31.17
A PID cannot be weakened to a UFD here. Let $A = k[x, y]_{(x,y)}$*, a local UFD with maximal ideal* $\mathfrak{m} = (x, y)$*. The elements* x, y *have no common prime factor, but* $(x, y) = \mathfrak{m} \neq A$*, and both lie in* $\mathfrak{m}$*, so no common factor* d *can be removed to leave an element outside* $\mathfrak{m}$*. The next lemma genuinely fails for such rings: for* $k = \mathbb{R}$*,* $B = A[y/x]$*,* $\mathfrak{p} = (x, y)$ *and* $\mathfrak{b} = (x, (y/x)^2 + 1)$ *we have* $\mathfrak{p}B \subseteq \mathfrak{b}$ *and* $B/\mathfrak{b} \cong \mathbb{C}$*, which has degree 2 over* $A/\mathfrak{p} = \mathbb{R}$*, although* B *and* A *have the same quotient field.*

Lemma 3.31.18
Let L/K *be a finite field extension,* A *a subring with quotient field* K *and* $B \supset A$ *an overring with quotient field* L*. Suppose that* $\mathfrak{p}$ *is a maximal ideal of* A *with* $A_{\mathfrak{p}}$ *a PID, and* $\mathfrak{b}$ *is a proper ideal of* B *containing* $\mathfrak{p}$*. Then canonically* $B/\mathfrak{b}$ *is an* $A/\mathfrak{p}$*-module and*

$$[B/\mathfrak{b} : A/\mathfrak{p}] \leq [L : K]$$

Furthermore we have $\mathfrak{b}B_{\mathfrak{p}} \cap A_{\mathfrak{p}} = \mathfrak{p}A_{\mathfrak{p}}$ *and*

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$$[B/\mathfrak{b} : A/\mathfrak{p}] = \left[B_\mathfrak{p}/\mathfrak{b}B_\mathfrak{p} : A_\mathfrak{p}/\mathfrak{p}A_\mathfrak{p}\right]$$

Proof. Suppose that $\overline{x_1}, ..., \overline{x_n} \in B/\mathfrak{b}$ are linearly independent over $A/\mathfrak{p}$. It is sufficient to show that $x_1, ..., x_n \in B$ are linearly independent over K, as then $n \leq [L : K]$, and the $\overline{x_i}$ were arbitrary linearly independent elements. Suppose we have a linear dependence

$$0 = \sum_{i=1}^{n} \lambda_i x_i \quad \lambda_i \in K$$

Then by clearing denominators we may assume that $\lambda_i \in A_\mathfrak{p}$ not all zero. By (3.31.16) (applied to the local PID $A_\mathfrak{p}$) we may divide the λ_i by a common non-zero factor, which does not affect the dependence, and assume that at least one λ_i is not in $\mathfrak{p}A_\mathfrak{p}$. Multiplying by a common denominator $s \in A \setminus \mathfrak{p}$ (a unit of $A_\mathfrak{p}$) we may take $\lambda_i \in A$, and some λ_i is still not in $\mathfrak{p}A_\mathfrak{p}$, so $\lambda_i \notin \mathfrak{p}$ (as $\mathfrak{p} \subseteq \mathfrak{p}A_\mathfrak{p}$). Reducing the dependence modulo $\mathfrak{b}$ gives $\sum_i \overline{\lambda_i}\overline{x_i} = 0$ in $B/\mathfrak{b}$ with $\overline{\lambda_i} \in A/\mathfrak{p}$ not all zero, contradicting the linear independence of the $\overline{x_i}$.

For the last two statements note that $\mathfrak{b} \cap A$ is a proper ideal containing the maximal ideal $\mathfrak{p}$, so $\mathfrak{b} \cap A = \mathfrak{p}$. Hence $\mathfrak{b}B_\mathfrak{p}$ is proper: if $1 = b'/t$ with $b' \in \mathfrak{b}$, $t \in A \setminus \mathfrak{p}$ then $t = b' \in \mathfrak{b} \cap A = \mathfrak{p}$, a contradiction. So $\mathfrak{b}B_\mathfrak{p} \cap A_\mathfrak{p}$ is a proper ideal of the local ring $A_\mathfrak{p}$ containing $\mathfrak{p}A_\mathfrak{p}$, hence equal to it. For the last statement consider the commutative diagram

$$\begin{array}{ccc} A/\mathfrak{p} & \hookrightarrow & B/\mathfrak{b} \\ \downarrow \sim & & \downarrow \sim \\ A_\mathfrak{p}/\mathfrak{p}A_\mathfrak{p} & \hookrightarrow & B_\mathfrak{p}/\mathfrak{b}B_\mathfrak{p} \end{array}$$

Both vertical arrows are isomorphisms by (3.5.26), applied with $S := A \setminus \mathfrak{p}$, as S is co-prime to the ideal in each case: for $s \in S$ we have $As + \mathfrak{p} = A$ by maximality of $\mathfrak{p}$, whence $Bs + \mathfrak{p}B = B$ and so $Bs + \mathfrak{b} = B$, as $\mathfrak{p}B \subseteq \mathfrak{b}$. So the diagram identifies the extension $A/\mathfrak{p} \subseteq B/\mathfrak{b}$ with $A_\mathfrak{p}/\mathfrak{p}A_\mathfrak{p} \subseteq B_\mathfrak{p}/\mathfrak{b}B_\mathfrak{p}$, which gives the equality of the degrees. ∎

Lemma 3.31.19 (Local Form of Ramification Index / Inertial Degree)
Using the notation in (3.31.12), suppose $\mathfrak{p}$ is a non-zero prime ideal and define $A' := A_\mathfrak{p}$, $\mathfrak{p}' := \mathfrak{p}A_\mathfrak{p}$, $B' := B_\mathfrak{p}$. Then A', B' is also "AKLB" and there is a bijection

$$\{\mathfrak{q} \lhd B : \mathfrak{q} \mid \mathfrak{p}\} \longleftrightarrow \{\mathfrak{q}' \lhd B' : \mathfrak{q}' \mid \mathfrak{p}'\}$$

under which the ramification index and inertial degrees agree, i.e.

$$e_{\mathfrak{q}/\mathfrak{p}} = e_{\mathfrak{q}'/\mathfrak{p}'}$$

and

$$f_{\mathfrak{q}/\mathfrak{p}} = f_{\mathfrak{q}'/\mathfrak{p}'} \leq [L : K]$$

Further the ramification index is equal to

$$v_{\mathfrak{q}'}(\pi B')$$

where π is any uniformiser for the DVR A'.

Proof. Evidently A' has quotient field K and B' has quotient field L. A' is a DVR (3.31.1) and therefore a Dedekind domain (3.31.3). By (3.23.9) B' is the integral closure of A', and therefore integrally closed. It has Krull Dimension 1 by (3.23.22) and, B being Noetherian (Dedekind, (3.31.1)), $B' = B_\mathfrak{p}$ is Noetherian too (3.14.7). Therefore we have the same situation as (3.31.12).

First consider the commutative diagram

$$\begin{array}{ccc} A & \hookrightarrow & B \\ \downarrow & & \downarrow \\ A' & \hookrightarrow & B' \end{array}$$

regarded as subrings of K and L respectively.

Then by chasing the diagram $\mathfrak{p}'B' = (\mathfrak{p}B)B'$. By (3.31.14) we have a unique factorisation

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$$\mathfrak{p}B = \prod_{i=1}^{n} \mathfrak{q}_i^{e_i}$$

for prime ideals $\mathfrak{q}_i$ of B dividing $\mathfrak{p}$. Define $\mathfrak{q}_i' := \mathfrak{q}_i B_{\mathfrak{p}}$ then these are prime by (3.5.18) (and the fact $\mathfrak{q} \cap (A \setminus \mathfrak{p}) = \emptyset$) and contain $\mathfrak{p}'B'$ (3.31.14). Then by (3.3.51)

$$\mathfrak{p}'B' = \prod_{i=1}^{n} (\mathfrak{q}_i')^{e_i}$$

The bijection of prime ideals is well-defined by (3.5.19).

By uniqueness $e_{\mathfrak{q}'/\mathfrak{p}'} = e_{\mathfrak{q}/\mathfrak{p}}$ for all primes $\mathfrak{q} \mid \mathfrak{p}$.

The relation $f_{\mathfrak{q}'/\mathfrak{p}'} = f_{\mathfrak{q}/\mathfrak{p}} \leq [L : K]$ follows directly from (3.31.18). ∎

Proposition 3.31.20 (Ramification Formula)
Using the notation of (3.31.12). For every non-zero prime ideal $\mathfrak{p}$ of A we have the identity

$$\sum_{\mathfrak{q}|\mathfrak{p}} e_{\mathfrak{q}/\mathfrak{p}} f_{\mathfrak{q}/\mathfrak{p}} = [B/\mathfrak{p}B : A/\mathfrak{p}] \leq [L : K]$$

Suppose in addition that B is a finite A-module then we have equality. In particular this holds when L/K is separable.

Proof. Recall from (3.31.14) we have the factorisation

$$\mathfrak{p}B = \prod_{\mathfrak{q}|\mathfrak{p}} \mathfrak{q}^{e_{\mathfrak{q}/\mathfrak{p}}}$$

By the Chinese Remainder Theorem (3.3.63) we have a canonical isomorphism

$$B/\mathfrak{p}B \xrightarrow{\sim} \prod_{\mathfrak{q}|\mathfrak{p}} B/\mathfrak{q}^{e_{\mathfrak{q}}}$$

Both $B/\mathfrak{p}B$ and $B/\mathfrak{q}^{e_{\mathfrak{q}}}$ are $A/\mathfrak{p}$-modules, and so equating vector space dimensions we have

$$[B/\mathfrak{p}B : A/\mathfrak{p}] = \sum_{\mathfrak{q}|\mathfrak{p}} [B/\mathfrak{q}^{e_{\mathfrak{q}}} : A/\mathfrak{p}]$$

As $\mathfrak{p}B \subset \mathfrak{q}^e$ we have a decreasing sequence of $A/\mathfrak{p}$-submodules

$$B/\mathfrak{q}^e \supseteq \mathfrak{q}^1/\mathfrak{q}^e \supseteq \ldots \supseteq \mathfrak{q}^e/\mathfrak{q}^e = (0)$$

so by (3.3.131) we have

$$[B/\mathfrak{q}^e : A/\mathfrak{p}] = \sum_{i=1}^{e} [(\mathfrak{q}^{i-1}/\mathfrak{q}^e)/(\mathfrak{q}^i/\mathfrak{q}^e) : A/\mathfrak{p}]$$

We claim there is an $A/\mathfrak{p}$-module isomorphism

$$(\mathfrak{q}^{i-1}/\mathfrak{q}^e)/(\mathfrak{q}^i/\mathfrak{q}^e) \to \mathfrak{q}^{i-1}/\mathfrak{q}^i$$

This exists as an A-module isomorphism by (3.3.106), and therefore as an $A/\mathfrak{p}$-module isomorphism as the action factors through the quotient map $A \to A/\mathfrak{p}$. Therefore by (3.18.15)

$$\begin{aligned}
[B/\mathfrak{q}^e : A/\mathfrak{p}] &= \sum_{i=1}^{e} [\mathfrak{q}^{i-1}/\mathfrak{q}^i : A/\mathfrak{p}] \\
&= \sum_{i=1}^{e} [B/\mathfrak{q} : A/\mathfrak{p}][\mathfrak{q}^{i-1}/\mathfrak{q}^i : B/\mathfrak{q}] \\
&= \sum_{i=1}^{e} f_{\mathfrak{q}/\mathfrak{p}} [\mathfrak{q}^{i-1}/\mathfrak{q}^i : B/\mathfrak{q}]
\end{aligned}$$

We claim that $[\mathfrak{q}^{i-1}/\mathfrak{q}^i : B/\mathfrak{q}] = 1$. This quotient is non-zero, as $v_{\mathfrak{q}}(\mathfrak{q}^{i-1}) = i - 1 \neq i = v_{\mathfrak{q}}(\mathfrak{q}^i)$ (3.31.9) shows $\mathfrak{q}^{i-1} \neq \mathfrak{q}^i$. Any subspace of $\mathfrak{q}^{i-1}/\mathfrak{q}^i$ other than 0 and the whole space is a-fortiori a B-submodule, and by the isomorphism theorem yields an ideal strictly between $\mathfrak{q}^{i-1}$ and $\mathfrak{q}^i$. Therefore it is enough to show there is no such ideal. By (3.5.42) we may localize at $\mathfrak{q}$ to consider the ring $B_{\mathfrak{q}}$. This is a DVR (3.31.1) and the result follows from (3.30.2). Therefore

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$$[B/\mathfrak{p}B : A/\mathfrak{p}] = \sum_{\mathfrak{q}|\mathfrak{p}} e_{\mathfrak{q}/\mathfrak{p}} f_{\mathfrak{q}/\mathfrak{p}}.$$

The inequality follows from (3.31.18) applied with $\mathfrak{b} = \mathfrak{p}B$ (a proper ideal containing $\mathfrak{p}$, as $\mathfrak{p}B \subseteq \mathfrak{q}$), where $\mathfrak{p}$ is maximal and $A_{\mathfrak{p}}$ a DVR as A is Dedekind and $\mathfrak{p} \neq 0$.

It remains to show this equals $[B/\mathfrak{p}B : A/\mathfrak{p}] = [L : K]$ in the case B is a finite A-module. By (3.31.19) and the last statement of (3.31.18) (applied with $\mathfrak{b} = \mathfrak{p}B$) we may replace A, B by $A_{\mathfrak{p}}, B_{\mathfrak{p}}$, where $B_{\mathfrak{p}}$ is again a finite $A_{\mathfrak{p}}$-module, and so reduce to the case of A a discrete valuation ring, in particular a principal ideal domain. Then by (3.23.25) B is a free A-module of rank $n = [L : K]$, so $B/\mathfrak{p}B \cong (A/\mathfrak{p})^n$ and $[B/\mathfrak{p}B : A/\mathfrak{p}] = n = [L : K]$.

The case L/K is separable is then (3.23.26). ∎

3.31.1 Local Monogenicity

Lemma 3.31.21 (A Uniformiser Generates a Totally Ramified Extension)
*Using the notation of (3.31.12), with L/K separable, let $\mathfrak{p}$ be a non-zero prime of A with a **unique** prime $\mathfrak{q}$ of B over it, of residue degree $f_{\mathfrak{q}/\mathfrak{p}} = 1$ (so $B_{\mathfrak{p}} = B_{\mathfrak{q}}$ is itself a DVR, totally ramified: $e_{\mathfrak{q}/\mathfrak{p}} = n := [L : K]$). Then for any uniformiser π of $B_{\mathfrak{p}}$,*

$$B_{\mathfrak{p}} = A_{\mathfrak{p}}[\pi],$$

and π's minimal polynomial $F \in A_{\mathfrak{p}}[Y]$ over K has degree exactly n.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

For $0 \leq j < n$, $\mathfrak{q}^j/\mathfrak{q}^{j+1}$ is 1-dimensional over $B_{\mathfrak{p}}/\mathfrak{q} = k$ (as $B_{\mathfrak{p}}$ is a DVR), generated by the image of π^j; residue degree 1 means $A_{\mathfrak{p}}/\mathfrak{p} = k$ too, so by downward induction on j, $\{1, \pi, ..., \pi^{n-1}\}$ spans $B_{\mathfrak{p}}/\mathfrak{p}B_{\mathfrak{p}} = B_{\mathfrak{p}}/\mathfrak{q}^n$ (as $\mathfrak{p}B_{\mathfrak{p}} = \mathfrak{q}^n$, totally ramified) as an $A_{\mathfrak{p}}/\mathfrak{p}$-module. By (3.20.6), applied to the finite $A_{\mathfrak{p}}$-module $B_{\mathfrak{p}}$, a spanning set of $B_{\mathfrak{p}}/\mathfrak{p}B_{\mathfrak{p}}$ already spans $B_{\mathfrak{p}}$ itself, so

$$B_{\mathfrak{p}} = A_{\mathfrak{p}}[\pi].$$

Let F be π's minimal polynomial over K, of degree d. Then $F \in A_{\mathfrak{p}}[Y]$ by (3.23.18), as $\pi \in B_{\mathfrak{p}}$ is integral over the integrally closed domain $A_{\mathfrak{p}}$ (3.23.9). As $B_{\mathfrak{p}} = A_{\mathfrak{p}}[\pi]$ and $F(\pi) = 0$, $B_{\mathfrak{p}}$ is already spanned by $1, \pi, ..., \pi^{d-1}$ over $A_{\mathfrak{p}}$, so $\operatorname{rank}_{A_{\mathfrak{p}}} B_{\mathfrak{p}} \leq d$; as $B_{\mathfrak{p}}$ is $A_{\mathfrak{p}}$-free of rank n (3.23.25), $d \geq n$, while $d = [K(\pi) : K] \leq [L : K] = n$ as $K(\pi) \subseteq L$. So $d = n$ exactly. ∎

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